Bryan Wei

LinkedIn GitHub Portfolio l76wei@uwaterloo.ca

Education

University of Waterloo, *Honours Computer Science Co-op* *Sep 2024 – Apr 2029*

Experience

Full Stack Developer *Sep 2026 – Dec 2026*
Workplace Safety and Insurance Board – Innovation Lab *Waterloo, ON*

- Engineered a **multithreaded synthetic test-data engine** that writes schema-configurable datasets to SQL files and Microsoft Fabric tables, cutting 1M-row generation time by **65% (20s → 7s)**
- Built an **indoor wayfinding app** helping **3,500+ employees** across **8 offices** learn their workspaces during return-to-office, using QR-anchored positioning and **Dijkstra routing** over multi-floor maps

Software Developer *Sep 2026 – Present*
Waterloo Reality Labs *Waterloo, ON*

- Developed a **few-shot Siamese network** using **PyTorch triplet loss** that recognizes new hand gestures from only 1–3 reference clips at up to **98% accuracy**, deployed via **ONNX** for in-engine inference in **Unity**
- Designed a **gesture intensity scoring module** that rates hand-gesture effort on a normalized 0–1 scale from kinematic features (joint velocity, snap, movement time), achieving **93% ranking accuracy** across effort levels

Software Developer *May 2026 – Aug 2026*
Traces *Toronto, ON*

- Built core features for Traces, a **developer platform** used by **100+ organizations** to capture, share, and continue AI coding-agent sessions across tools and teams
- Implemented **data ingestion adapters** for **14+ coding agents**, handling local session discovery, content parsing, and **schema normalization** across JSONL files, SQLite databases, and cloud-backed session stores
- Engineered a **subagent discovery pipeline** that infers parent–child relationships for **8+ agents** with **97% accuracy**, preserving lineage across local indexing, API surfaces, remote sync, and **Convex** storage
- Developed **token usage tracking** for **8+ agents** at **99% accuracy**, mapping agent-specific per-API-call and session-level usage into shared schemas, with CLI sync and persistence in **Convex** and **Tinybird**
- Designed a provider-neutral **cloud agent session capture engine** spanning polling, webhook, and streaming interfaces, with **concurrent job processing** and **idempotent, retry-safe persistence**

Projects

StickDrift | GitHub Demo Devpost *Oct 2026*

- Developed a **movement disorder monitoring platform** for seniors, where senior-oriented controller games capture joystick and accelerometer data to detect early signs of movement disorders with machine learning
- Trained a **PyTorch Temporal Convolutional Network** on joystick and accelerometer time series to detect tremor, stroke, and Parkinson's movement patterns with **96% accuracy**, feeding pattern scores into a caregiver dashboard
- Built a **telemetry pipeline** streaming controller input over **FastAPI WebSockets** for live caregiver monitoring

Contour | GitHub Live *Sep 2026*

- Built a **project visualization platform** that automatically transforms teammates' GitHub activity and coding-agent sessions into one interactive project map, piecing together work scattered across tools and devices
- Engineered **GitHub webhook** ingestion for PRs, reviews, and CI runs, plus a local collector that **incrementally syncs** coding-agent sessions via **SQLite** checkpoints, cutting re-uploads by **99%** and making rescans **3.6× faster**
- Designed an **asynchronous event-processing pipeline** on **pg-boss** that serializes processing per project, merging concurrent webhook and local-collector events into the live map organized by the **OpenAI API**, with **0% duplicates**

Skills

Languages: Python, TypeScript, JavaScript, Java, C#, C++, C, SQL, Bash, PowerShell

Frameworks: React, Next.js, React Native, three.js, Node.js, Express, FastAPI, Spring Boot, PyTorch

Tools & Databases: Git, Linux, AWS, Azure, Unity, Bun, PostgreSQL, MySQL, SQLite, MongoDB, Convex, Tinybird